recommend C." You had better take their advice! What they are really saying is: *"Our DSP is so difficult to program in assembly that you will need 6 months of training to use it."* On the other hand, some DSPs are easy to program in assembly. For instance, the Analog Devices products are in this category. Just ask their engineers; they are very proud of this.

One of the best ways to make decisions about DSP products and software is to speak with engineers who have used them. Ask the manufacturers for references of companies using their products, or search the web for people you can e-mail. Don't be shy; engineers love to give their opinions on products they have used. They will be flattered that you asked.

How Fast are DSPs?

The primary reason for using a DSP instead of a traditional microprocessor is *speed*, the ability to move samples into the device, carry out the needed mathematical operations, and output the processed data. This brings up the question: How fast are DSPs? The usual way of answering this question is **benchmarks**, methods for expressing the speed of a microprocessor as a number. For instance, fixed point systems are often quoted in **MIPS** (million integer operations per second). Likewise, floating point devices can be specified in **MFLOPS** (million floating point operations per second).

One hundred and fifty years ago, British Prime Minister Benjamin Disraeli declared that there are three types of lies: *lies*, *damn lies*, and *statistics*. If Disraeli were alive today and working with microprocessors, he would add *benchmarks* as a fourth category. The idea behind benchmarks is to provide a head-to-head comparison to show which is the best device. Unfortunately, this often fails in practicality, because different microprocessors excel in different areas. Imagine asking the question: Which is the better car, a Cadillac or a Ferrari? It depends on what you want it for!

Confusion about benchmarks is aggravated by the competitive nature of the electronics industry. Manufacturers want to show their products in the best light, and they will use any ambiguity in the testing procedure to their advantage. There is an old saying in electronics: "*A specification writer can get twice as much performance from a device as an engineer*." These people aren't being untruthful, they are just paid to have good imaginations. Benchmarks should be viewed as a *tool* for a complicated task. If you are inexperienced in using this tool, you may come to the wrong conclusion. A better approach is to look for specific information on the execution speed of the algorithms you plan to carry out. For instance, if your application calls for an FIR filter, look for the exact number of clock cycles it takes for the device to execute this particular task.

Using this strategy, let's look at the time required to execute various algorithms on our featured DSP, the Analog Devices SHARC family. Keep